

Customer Feedback Analysis at Scale

Florian Ernst - 2024

Introduction

About me:

Name: Florian Ernst (@mySugr for 8 years)

(iOS) Mobile Developer -> Data Engineer -> Security Eng.

About what I'm building:

- Is in an early alpha phase & under development.
- Result of many weekends talking to Large Language Models.
- Still in POC phase.

Agenda

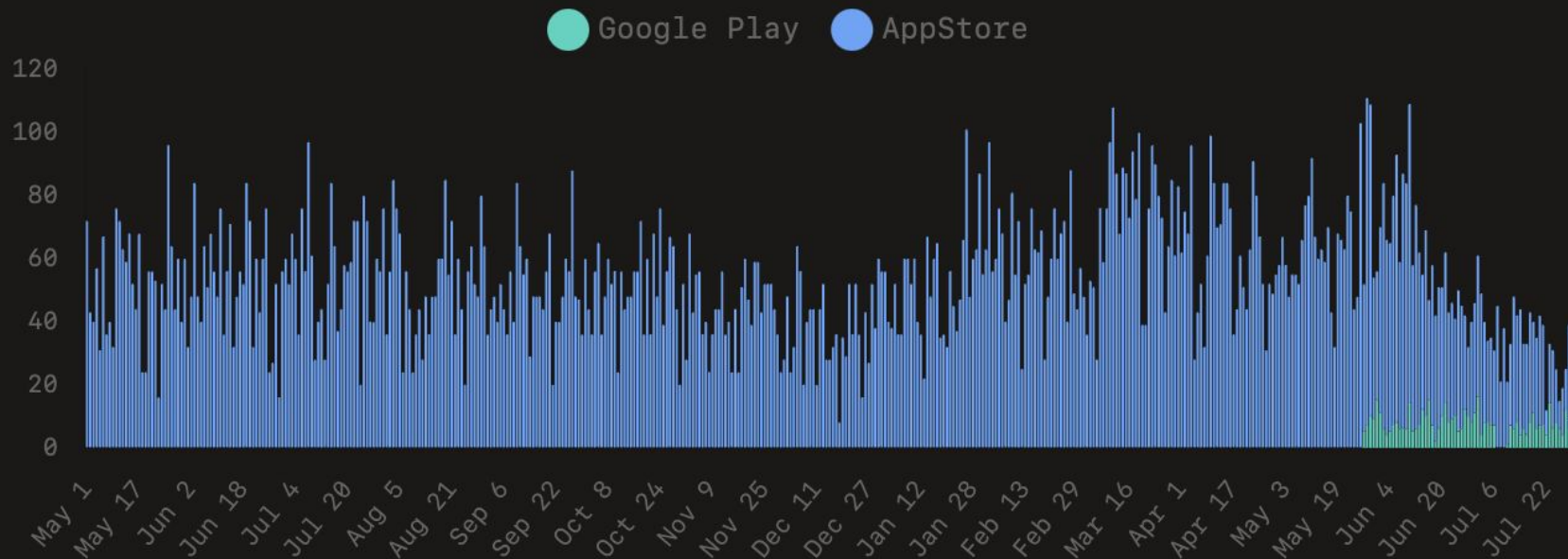
- The Problem
- The Goal
- The Solution?
- Feature Overview
- How does it work?
- Miscellaneous
- Demo Time

Feel free to raise your hand at any time for questions 🙋

The Problem

- Analyzing large volumes of diverse customer feedback is resource-intensive.
- Manual analysis methods are slow, biased, and miss contextual nuances.
- Delayed feedback processing hinders timely action on emerging issues.
- Tracking metrics like downloads and ratings alone fails to capture the full spectrum of customer sentiment and insights.

Incoming Feedback by Data Source



The Goal

- Making sense of customer feedback.
- Providing tools to arrive at actionable insights.
 - Summaries, Reports, Dashboards, Search Engines..
- Identifying in-the-wild issues early.
- Help with the detection of special information in customer feedback (safety relevant).
- Use the same approach to identify weaknesses and areas for improvement in competitor products.

The Solution?

LLMs are trained on trillions of tokens of human written text, which makes them good at:

- Natural Language Processing
- Sentiment detection
- Context understanding
- Filtering of noise
- Extraction of features

Conclusion: Use LLMs to analyse the feedback, categorize it & create summaries and insights.

Feature Overview

Default Feature Extraction

- Positive & Negative Keywords
- Issues / Bugs
- Feature Requests
- Questions
- Sentiment

huw_nant via AppStore from 🇬🇧 United Kingdom 4/5 ★ May 25, 2024 08:50

simple nice and clear; flexible and helpful, graphics a little bit much; bluetooth connection doesn't always work (Accu-Chek Instant) so need to enter manually sometimes; works seamlessly with Apple Health

Tags: 😊 Mixed 🐛 Bug Report

Positive: Navigation 3rd Party Integrations

Negative: Bluetooth Design

Feedback Summarization



Most mentioned Issues

119

46

The estimated HbA1c/Estimated A1C feature is no longer displayed on the app following recent updates, despite users having the necessary data.

Countries —



United States



Austria



Argentina



United Kingdom



Japan



Brazil



Canada



Switzerland



Netherlands



Belgium



France



Australia



India



Germany



Norway



Saudi Arabia



Finland



Italy



Mexico

19

Users encounter connectivity problems with the meter or other health devices, either failing to connect at all or having intermittent connections.

Countries ▼

Feedback Summarization

★ Most requested Features

77

15

Features requested include integration with devices like blood pressure monitors, fitness apps, and direct integration with health applications like Apple Health.

Countries ▼

12

Requests include customizable fields for blood glucose logging, meal logging customization, adding fields for meal impacts, and user-customizable food and tags.

Countries ▼

Feedback Summarization

? Most asked Questions

40

33

What happened to the HbA1c/A1C display and when will it be restored?

Countries ▼

7

How does one change settings or understand features to use the app effectively?

Countries ▼

Custom Tags

Name

 Safety Relevant

 Save Changes

Delete

Cancel

Ruleset

```
[*] If the review mentions severe hypoglycemia or hyperglycemia caused by therapy decisions based on the app.
```

Color

Yellow



 Testing

```
The bolus calculator returned wrong values
```

 Test Rules

Tag Applied:  Yes

Analytics & Filters

Countries

Germany x

Data Sources

AppStore x

Sentiment

👎 Negative x

Tags

Select Tags

Custom Tags

Select Custom Tags

Keywords

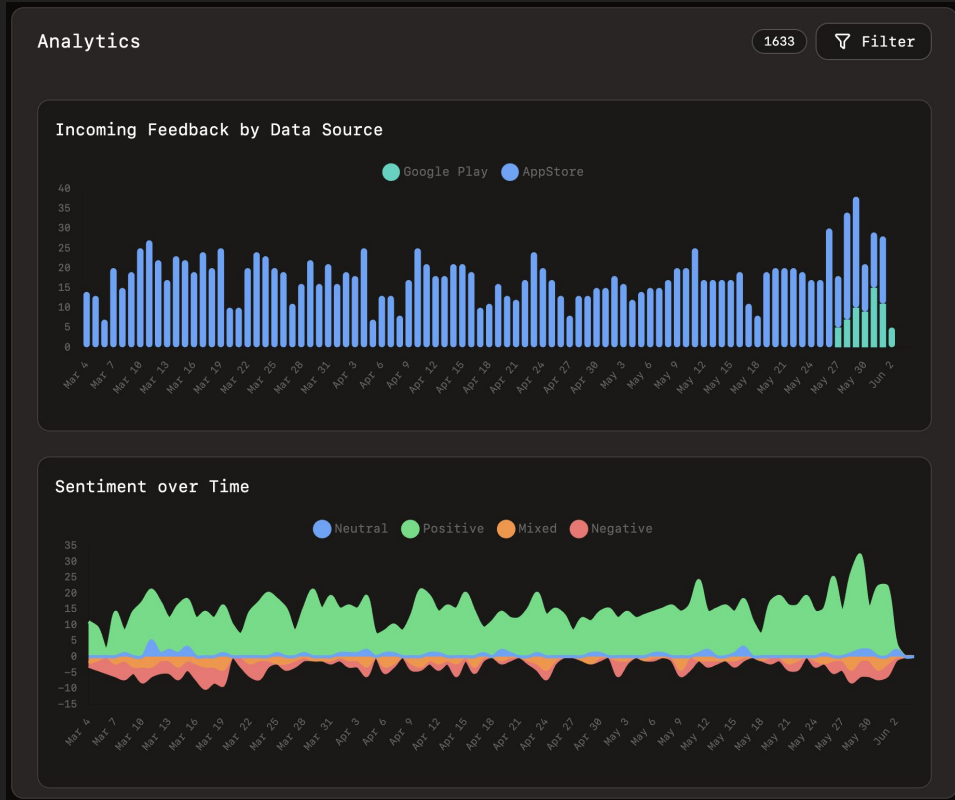
Select keywords

Date Range

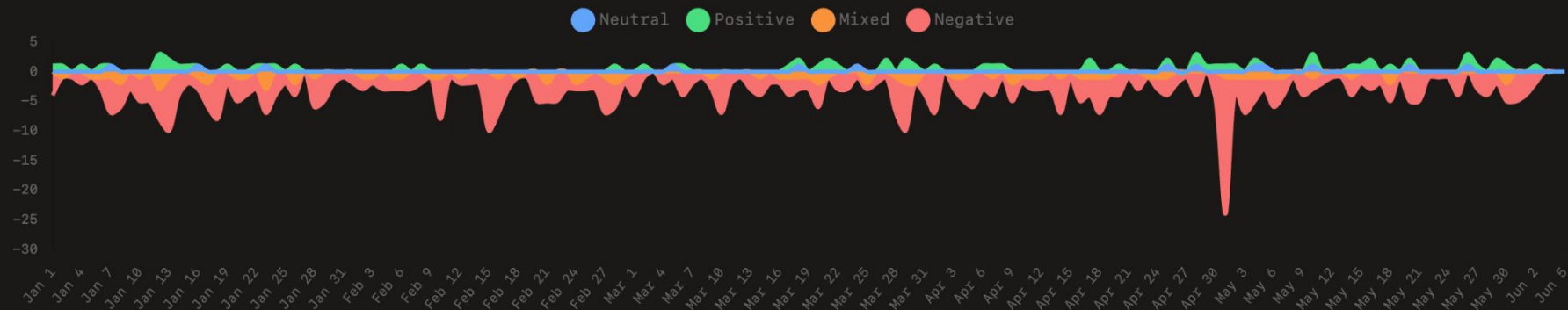
03/04/2024

 to

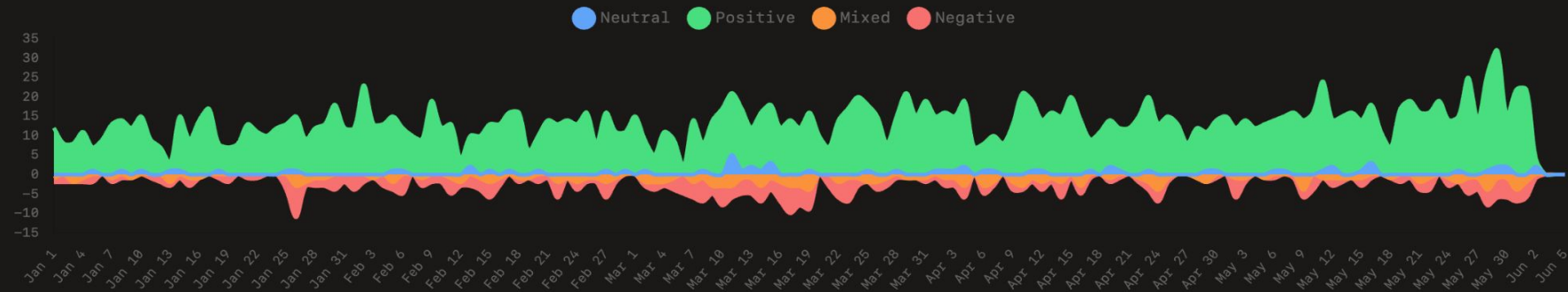
06/04/2024



Sentiment over Time



Sentiment over Time



Full Text & Semantic search

Search

Search Query

Sort by

Type

Date

▼

Semantic

▼

🔍

 Search

385 results found.

Ganesh Assisi via

AppStore

 from

🇮🇳 India

5/5 ⭐

 Aug 5, 2023 01:53

Easy to use User friendly interface

Tags:

😊 Positive

Positive:

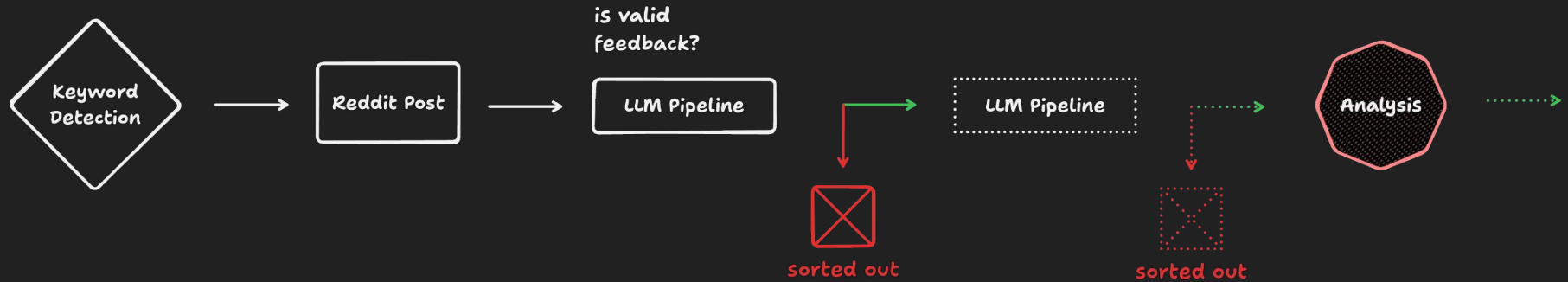
Design

Search Score: 0.567

Supported Data Sources

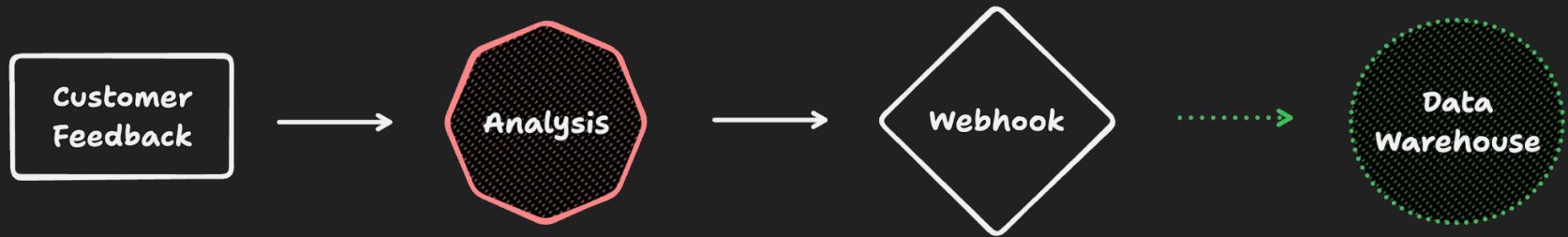
- Apple's AppStore (API / Scraper)
- Google Play (API / CSV / Scraper)
- Rest API
- Reddit Live Monitoring
- More to come...

LLM powered Web Scrapping



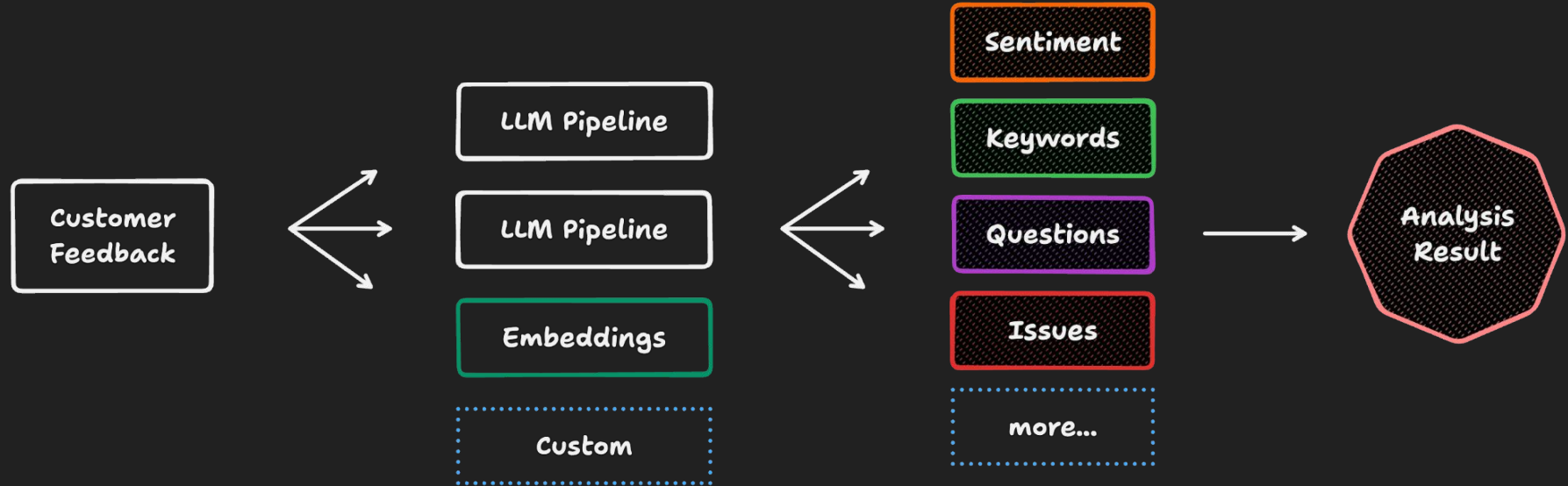
Webhook Mode

In this mode, a webhook (JSON POST) is called once the analysis is finished and no data is stored in the database.

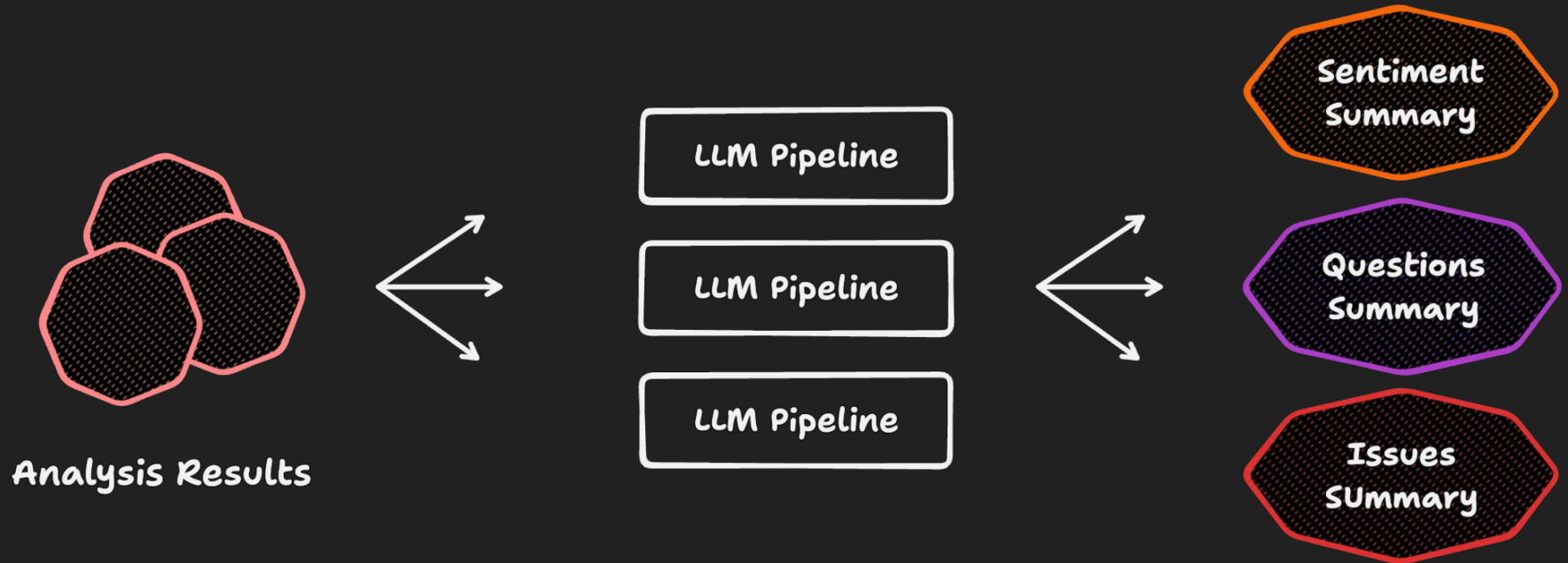


How does it work?

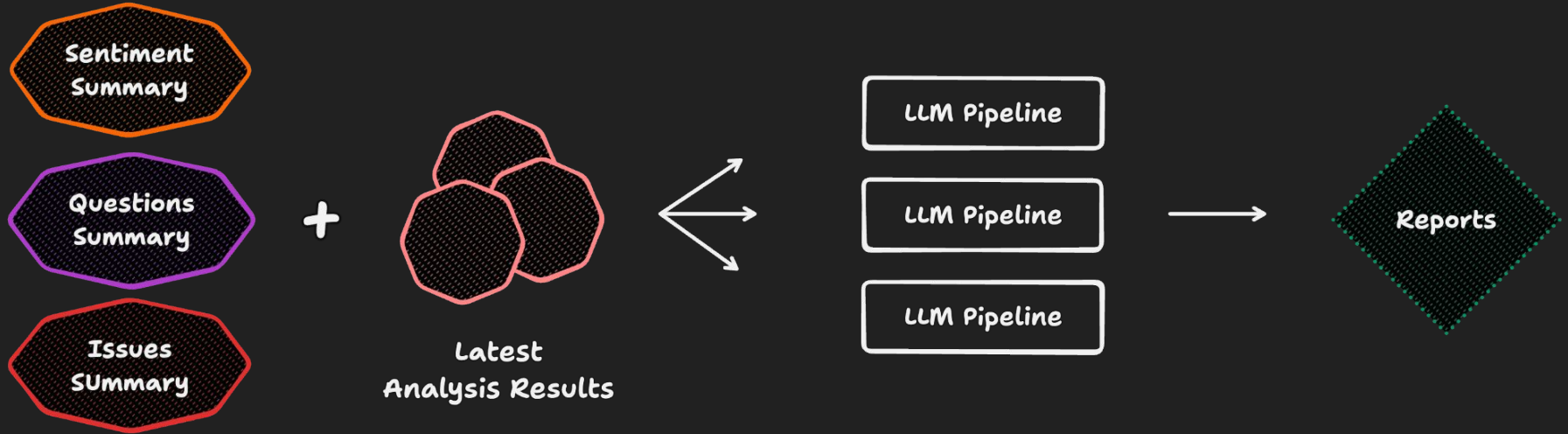
Step 1 - Analysing the Feedback



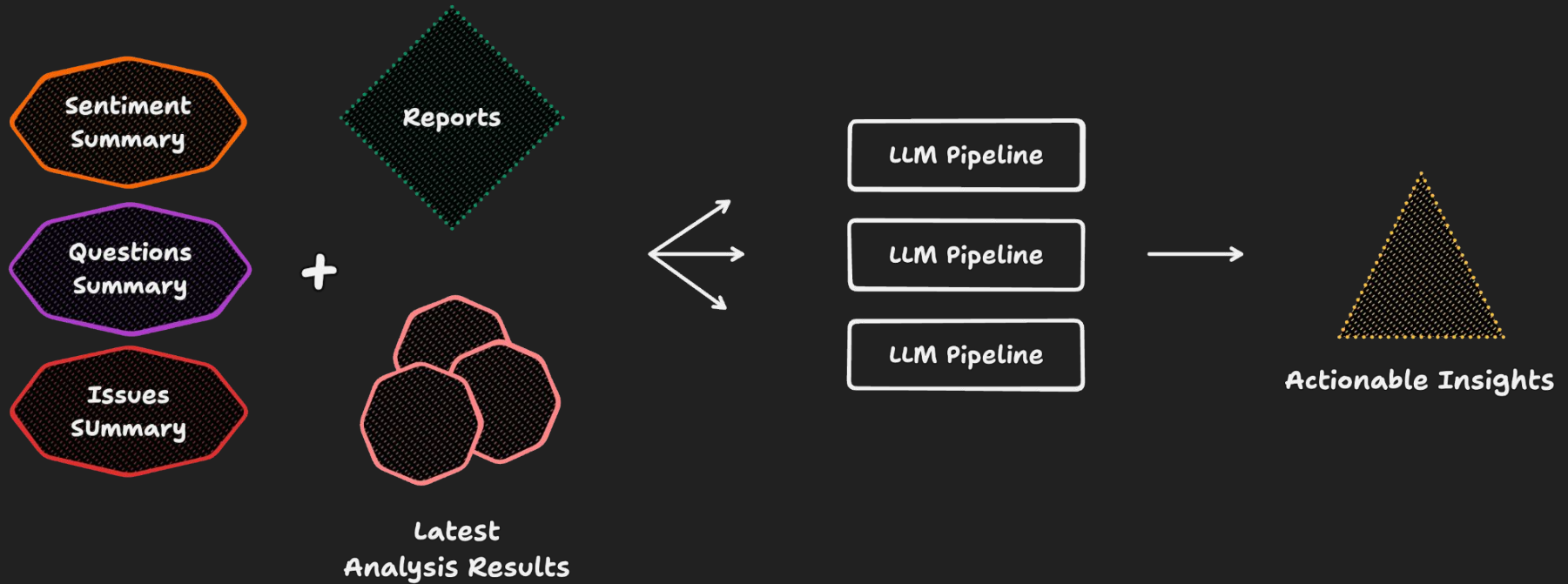
Step 2 - Creating Internal Summaries



Step 3 - Creating Reports



Step 4 - Creating Insights



Speed, Models & more

- Mix of Anthropic Claude, GPT-4o, llama3
 - Pipelines highly configurable.
- Written in Go, HTMX & Tailwind.
- Heavy lifting is outsourced to Serverless GPUs and LLM API providers.
- Could probably be moved completely on-premise.
- Demo runs on a 3.49 EUR / Month Server

Roadmap & Future Outlook

- More Data Sources
- Dashboards
- More analysis types
 - Mark app as competitor to run different pipelines
- ChatGPT like interface
- Cross App comparisons

This is the worst AI will ever be.

Demo Time